Omar Elarabi

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LinkedIn **Portfolio**

EDUCATION

Queen Mary University of London, Aerospace Engineering, 1st Class Honours **September 2023 - July 2026**

- ***Key Modules Completed:*** Designing for Sustainable Manufacture, Control System Analysis and Design, Flight Conceptual Design, Materials Engineering, Aircraft Propulsion Systems & Structures, Aerothermodynamics of Flows, High-Speed Flows.

PROFESSIONAL EXPERIENCE

Peer-Assisted Study Support (PASS) Student Organiser **August 2025 - June 2026**

- Monitored session schedules and room allocations using Excel-based analysis, checklists of essential requirements, and coordination with departments, whilst planning promotional activities and coordinating 10 mentors to deliver sessions.
- Maintained module page and feedback analytics, liaised with staff and professors to align sessions with student needs, and implemented improvements based on data-driven insights, increasing engagement by 40%.

Peer-Assisted Study Support (PASS) Mentor **September 2024 - June 2026**

- Recognised and addressed specific areas where students struggled, developing strategies to address these weaknesses and improve overall performance by an average of 60%.
- Customised teaching approaches and managed sessions based on individual student needs, fostering a supportive learning environment and driving academic progress. Awarded Mentor of the Year.

ACADEMIC PROJECTS

'Designing the Powertrain Exhaust System – Formula Student UK' – SolidWorks & ANSYS Fluent **September 2025 - June 2026**

- Researched sustainable materials and manufacturing processes to optimise performance, weight, and sustainability. Designed headers, collector, tailpipe, and muffler in CAD, ensuring fit and compliance with chassis and IMechE rules. Used CFD and thermal simulations to refine gas flow, scavenging, and heat management. Conducted lifecycle analysis to guide material selection and sustainable manufacturing. Validated performance, noise compliance, and compatibility with vehicle.

'Flow over a Wing Profile – StarCCM+' – CFD & Excel **November 2025 - December 2025**

- Simulated 2D slotted aerofoil flow at 25 m/s and 5° AoA using finite volume CFD, comparing 1st and 2nd order discretisation across multiple mesh settings to assess numerical convergence. Processed 2,312 surface data points across four aerofoil and flap surfaces in Excel, analysing pressure and total pressure distributions to evaluate aerodynamic performance, quantify simulation accuracy and identify a 2nd order solution that reduced total pressure error from 4.13% to 2.05%.

'Redesigning the A320 Flap Track Fairing – National Composites Centre' – CAD & ANSYS **February 2025 - March 2025**

- Screened over 4,000 materials in ANSYS EduPack and applied Ashby's method to select a composite for the A320 flap track fairing, meeting constraints of 3.4 kg mass, 1.66 mm deflection, and stability at 44,000 ft. Refined the material composition to maximise stiffness-to-density ratio and sustainability, defining prepreg layup and autoclave curing methods. Incorporated sustainable strategies such as downcycling analysis and precision cutting to minimise waste. Presented to NCC stakeholders.

EXTRA-CURRICULAR & LEADERSHIP ROLES

Aurora Initiative, Health & Safety Officer, QMUL **August 2025 - June 2026**

- Conducted safety inductions and pre-activity briefings for society members, developed accessible briefing materials, and led annual risk assessments to mitigate hazards across all events and engineering activities.
- Maintained compliance and attendance records, coordinated with other officers and departments to update safety protocols, and implemented preventative measures that ensured activities ran safely and in line with engineering standards.

Hyperlink, Braking Team Member, Queen Mary University of London **September 2024 - May 2025**

- Designed concepts using sketching and CAD, of magnetic braking methods for primary and secondary braking systems including use of eddy currents for magnetic damping and braking pads for frictional drag.
- Incorporated dimensions and material research into maximising efficiency for primary system, and methods to primary system, whilst considering health and safety, including power faults.

Second Year Aerospace Engineering Course Representative, QMUL **September 2024 - May 2025**

- Built a communication channel between a sum of 190 students and staff to improve student experience in partnership with QMUL, students and student unions.
- Raised awareness of issues in meetings, reporting feedback from meetings and recording turnouts after improvements.

SKILLS & INTERESTS

IT Skills: Computer-aided Design, MATLAB, Python, Abaqus, Microsoft Office, Computational Fluid Dynamics, StarCCM+, ANSYS.